Architecture & Implementation Report: Custom Cloud-Based VPN

Deploying a Zero-Trust Network using Tailscale and Oracle Cloud Infrastructure

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Domain Target: Network Infrastructure & Cloud Security

1. Executive Summary

This technical report details the architectural design, configuration protocol, and security posture of a custom Virtual Private Network (VPN). The solution is implemented utilizing Tailscale's WireGuard-based overlay network, strategically integrated with an Oracle Cloud Infrastructure (OCI) instance functioning as a dedicated exit node. This architecture ensures secure, encrypted remote access, facilitates advanced network diagnostics, and provides highly secure traffic routing for local development endpoints, including the primary Lenovo ThinkPad L480 workstation.

2. Architectural Overview

The network is engineered upon a zero-trust architecture model, explicitly moving away from vulnerable perimeter-based security paradigms. Rather than exposing open ports to the public internet, the infrastructure relies on authenticated peer-to-peer cryptographic handshakes.

- **The Control Plane:** Tailscale acts as the central coordinator, securely managing the distribution of public keys and the dynamic coordination of peer locations across the mesh network.
- **The Data Plane:** The WireGuard protocol handles the high-throughput, low-latency encryption of the actual network traffic traveling between nodes.
- **The Exit Node Gateway:** An Oracle Cloud instance is configured to route all internet-bound traffic originating from the client devices. This masks the local IP address and provides a consistent, secure egress point.

3. Core Infrastructure Components

Component	Specification & Role
Cloud Provider	Oracle Cloud Infrastructure (OCI) - Always Free Tier Compute
Overlay Network Protocol	Tailscale (WireGuard)
Primary Client Node	Lenovo ThinkPad L480 (Local Workstation)
DNS Resolution	MagicDNS for automated internal hostname routing

4. Implementation Protocol

4.1 Cloud Instance Provisioning & Hardening

The foundational phase involved provisioning an OCI compute instance. The environment was deployed with a minimal Linux distribution to reduce the potential attack surface. Strict security lists and ingress/egress firewall rules were established at the Virtual Cloud Network (VCN) level, explicitly permitting necessary outbound connections while blocking unauthorized external inbound traffic.

4.2 Exit Node Routing Configuration

To enable the Oracle server to act as a secure gateway for external internet traffic, IPv4 forwarding was activated within the kernel parameters. This required modifying the system configuration to enable `net.ipv4.ip_forward`. Subsequently, the Tailscale daemon was executed with specific flags to advertise the instance to the mesh network as an available exit node.

> **Strategic Advantage - Bypassing Local ISP Constraints:** Operating a dedicated offshore exit node allows for precise packet inspection and seamless network troubleshooting without interference from local ISP caching layers or DNS latency (e.g., bypassing regional propagation delays).

4.3 Client Integration & Subnet Routing

The local development machine was authenticated to the tailored mesh network. MagicDNS was initialized to permit seamless internal hostname resolution without requiring manual IP address tracking. The client device is configured to force all outgoing web traffic through the Oracle exit node when active, bypassing local network restrictions and ensuring end-to-end encryption across untrusted local Wi-Fi or cellular networks.

5. Security Posture Analysis

This implementation significantly hardens the network environment against common interception techniques and Man-in-the-Middle (MitM) attacks. By relying on WireGuard's modern cryptographic primitives—specifically the Noise protocol framework, Curve25519, ChaCha20, and Poly1305—data transit remains completely confidential and tamper-evident. Furthermore, because Tailscale utilizes NAT traversal techniques without requiring open inbound firewall ports on the client side, the risk of external port scanning and unauthorized network penetration is effectively mitigated.

6. Conclusion

The deployment of this customized Tailscale VPN architecture successfully provides a robust, highly secure, and efficiently manageable networking environment. It stands as a practical demonstration of modern infrastructure-as-code principles, perfectly suited to support ongoing cybersecurity analyses, Capture The Flag (CTF) platform hosting environments, and secure remote server administration.